

Jeongmin Lee

 jeongminl |  jeongmin |  {firstname}@kaist.ac.kr

SUMMARY

I am interested in **computer architecture and systems**, with a specific focus on **hardware-software codesign** and **accelerator architecture** for machine learning applications to improve performance and efficiency.

EXPERIENCES

Undergraduate Researcher, ARC Lab, Seoul National University Jun 2026 - Present

- Developed a lifetime-informed KV cache routing mechanism to reduce flash wear on High Bandwidth Flash when used in heterogeneous memory systems, extending flash lifetime by up to 3x in Agentic AI serving scenarios.
- Implemented an Agentic AI serving simulator and various KV cache routing mechanisms used for evaluation.

Undergraduate Researcher, Litz Lab, UC Santa Cruz Jan 2026 - Jun 2026

- Implemented a module in SystemVerilog to measure bandwidth, latency etc. of a novel STTRAM module installed on a BittWare FPGA.
- Implemented STTRAM exposure to user space through PCIe via MMIO and DMA.
- Investigated potential vulnerabilities and other characteristics of the STTRAM module, presenting findings in the annual CRSS Industrial Advisory Board Meeting.

Undergraduate Researcher, VIA Group, KAIST Jul 2025 - Dec 2025

- Implemented a cycle-level simulator in C++ for a sparse CNN accelerator, simulating its speedups against a naive computation approach.
- Implemented a runtime memory manager in C++ and CUDA to decrease total memory requirement of DNNs by appropriately offloading data from GPU memory to CPU memory.

Undergraduate Researcher, RIRO Lab, KAIST Feb 2025 - Jul 2025

- Developed a long-term horizon planning and replanning robotics module with 50+% replanning success rate when faced with action execution disruptions.
- Implemented multiple robotic planning and replanning baselines, including CAPE, ISRLLM, Inner-Monologue, ProgPrompt, Text2Reaction, DoReMi, and Zero-Shot Planner, for deployment in the AI2THOR simulator.

Analyst, Defense Security Agency Jul 2023 - Jan 2025

- Developed and implemented SIGINT algorithms.

PUBLICATIONS

Kim, Yeseung, Gyungjin Eo, Jisang Park, **Jeongmin Lee**, Jun Park, and Daehyung Park (2026). “Fore-Sight: Reactive Long-Horizon Task Execution with Autoregressive Planning and Monitoring”. In Submission.

Yang, Jaehoon, **Jeongmin Lee**, Haneul Park, Seung Yul Lee, Nam Sung Kim, and Jae W. Lee (2026). “Lachesis: Lifetime-Aware KV Cache Placement for Agent Serving across HBM and High-Bandwidth Flash”. In Submission.

EDUCATION

2021 - present	Korea Advanced Institute of Science and Technology B.S in Computer Science GPA: 4.17/4.3 (Current) <i>Relevant Coursework:</i> Systems Programming, Computer Organization, Operating Systems, Introduction to Artificial Intelligence, Machine Learning Basics and Applications.	Daejeon, South Korea
2026	University of California, Santa Cruz Exchange Programme GPA: 4.0/4.0 <i>Relevant Coursework:</i> Logic Design, Parallel and Concurrent Programming, Compiler Design, Embedded System Design.	California, USA
2015 - 2020	Anglo-Chinese School (Independent) International Baccalaureate Score: 45/45	Singapore

AWARDS AND SCHOLARSHIPS

2026	IPESK Next-Generation Engineer <i>Awarded to top students in nationwide research universities.</i>	South Korea
2025	National Science & Technology Scholarship <i>Awarded to top engineering and science students nationwide.</i>	South Korea
2025	Department Honors Scholarship (Valedictorian) <i>Awarded to the top student in each department per semester.</i>	KAIST
2025	Dean's List, School of Computing <i>Awarded to the top 3% of students in the School of Computing per semester.</i>	KAIST
2020	Outstanding Award, International Mathematical Modelling Challenge <i>Highest award in the challenge, given to the top 2 teams out of the 54 participating teams worldwide.</i>	World
2018	Korean Association in Singapore Scholarship <i>Awarded to top Korean students in Singapore.</i>	Singapore
2018	First Place & POSB Award, i.code <i>National Infocomm Competition</i>	Singapore

SKILLS

Programming Languages	Python, C/C++, Rust, Verilog/SystemVerilog, Scala, Java.
Machine Learning Frameworks	PyTorch, TensorFlow.
Design Tools	Vivado.
Languages	Native: English, Korean. Limited: Spanish, Chinese.